Rich Text Editor, textQuestionEditor

25FEBS201IAE1

Seat No. : 05 FEB 2026

Student Name : .....

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, February 2026

ASH - SEMESTER II

25FEBS201 - Engineering Mathematics II

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

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Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	RBT Level
	a)For a complex number $-2i+3$ real part is _____ and imaginary part is _____.	1	CO1	L1
	b)Is $(e^x sin\ y)dx + (e^x cos\ y)dy = 0$, exact?	1	CO2	L1
	c)If $\frac{1}{2} \pm \frac{\sqrt{3}\,i}{2}$ are roots of second order differential equation, the complementary function is given by	1	CO3	L1
	d)State De'Moivre's theorem.	1	CO1	L1
	e)The solution of linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ is given by the formula _____.	1	CO2	L1
	f)If m is real and repeated root of third order differential equation then its complementary function is _____	1	CO3	L1
Q2	a)Apply **De Moivre's Theorem** to solve $z^5 = 1$.	5	CO1	L3
	OR			
	b)Apply De'Moivre's theorem to expand $cos^5\theta$ in series of cosines of multiples of θ.	5	CO1	L3
Q3	a)*Apply the rule of homogeneous differential equations and solve the given equation*	5	CO2	L3
	$(x^{\frac{4}{3}} + y^{\frac{4}{3}})dx - xy^3 dy = 0$ OR			

•	b)Apply the appropriate definition and necessary conditions to solve the given differential equation $(2x^2 + 3y^2 - 7)x\,dx + (3x^2 + 2y^2 - 8)y\,dy = 0$.	5	CO2	L3
• Q4	a)Solve $(D^2 - 2D + 1)y = x^2 + x$.	5	CO3	L3
	OR			
	b)Solve $(D^3 - 3D^2 + 3D - 1)y = xe^x + e^x$.	5	CO3	L3

